SIMATS ENGINEERING
ASSESSMENT TOOL 1
COURSE NAME: Artificial Intelligence COURSE CODE: CSA1741

CO1-AT1-Analytical Problem Solving

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COURSE OUTCOME COVERED

CO1: Apply AI basics such as intelligent agents and uninformed search methods to solve problems effectively. *(BL3)*

Assessment Tool Weightage: 40%

QUESTIONS: 5 **Total Marks: 50**

1. Solve the Water Jug problem: you are given 2 jugs, a 4-gallon one and 3-gallon one. Neither has any measuring maker on it. There is a pump that can be used to fill the jugs with water. How can you get exactly 2 gallons of water into 4-gallon jug? Explicit assumptions: A jug can be filled from the pump, water can be poured out of a jug onto the ground, water can be poured from one jug to another and that there are no other measuring devices available.

2. Find out how Mars Rover, analyse and explore the samples on Mars by answering the following questions.
   i. What are the percept's for this agent?
   ii. Characterize the operating environment.
   iii. What are the actions the agent can take?
   iv. How can one evaluate the performance of the agent?
   v. What sort of agent architecture do you think is most suitable for this agent? Why?

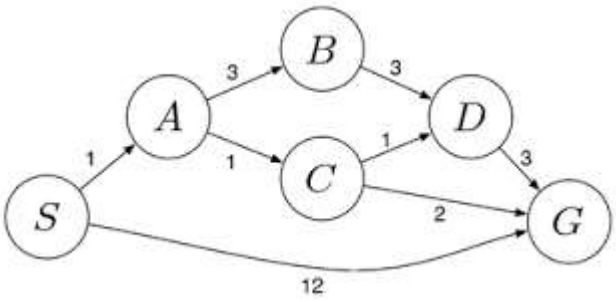
3. Explore the search space using search strategies and formulate the problem components for the 8-Queens problem using the following information: Place 8 queens on a chessboard such that none of the queen's attack any of the others. A configuration of 8 queens on the board as shown in the figure below, but this does not represent a solution as the queen in the first column is on the same diagonal as the queen in the last column.

4. A customer wants to travel from the one location to another location using OLA Cab booking mobile application. The customer can select the any of the cab type as mini, micro, sedan, shared, prime and so on based on his comfort to reach the destination. Identify what type of problem-solving agent can be used and write the pseudocode for the above problem.

5. A logistics company operates a delivery network where packages must be transported between warehouses at minimum cost. Each route between warehouses has an associated travel cost (fuel + time). The company wants to determine the least-cost

path from the starting warehouse **S** to the destination warehouse G using the Uniform Cost Search (UCS) algorithm.

The delivery network is represented as a weighted graph:



Code of Conduct:

I ___ D.N Shirshika _____ *(192425396) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.*
I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student: **D.N Shirshika**

RUBRICS FOR CALCULATION:

Criterion	Level 4 (Exemplary)	Level 3 (Proficient)	Level 2 (Developing)	Level 1 (Beginning)	Max Marks
Problem Understanding	Clearly interprets problem, identifies all parameters and requirements accurately	Understands problem with minor gaps	Partial understanding; misses key elements	Misinterprets or unable to understand problem	10
Method Selection	Selects most appropriate method/formula with strong justification	Selects correct method with minor errors	Inappropriate or partially correct method	Unable to select method	10
Solution Process	Logical, systematic steps with correct approach throughout	Mostly correct steps with minor mistakes	Disorganized steps; multiple errors	No clear process	12.5
Accuracy of Calculation	All calculations accurate; correct final answer	Minor calculation errors	Frequent errors; incorrect final answer	No valid calculations	10
Interpretation of Results	Clearly interprets results with units and engineering relevance	Basic interpretation with minor omissions	Limited or unclear interpretation	No interpretation	7.5

ANSWERS

1) Water Jug Problem'

Problem: Obtain exactly 2 gallons of water in a 4-gallon jug using a 4-gallon jug and a 3-gallon jug without any measuring instrument

Problem formulation

- Initial state: Both jugs are empty (0, 0).
- Goal state: The 4-gallon jug contains exactly 2 gallons.
- State space: All possible water quantities in both jugs.
- Operators: Fill, empty, and pour water between jugs.

Solution steps

Step	Action	4-gallon jug	3-gallon jug
1	Fill the 4-gallon jug	4	0
2	Pour into the 3-gallon jug until full	1	3
3	Empty the 3-gallon jug	1	0
4	Pour the remaining water into the 3-gallon jug	0	1
5	Fill the 4-gallon jug again	4	1
6	Pour into the 3-gallon jug until full	2	3

Final answer: The 4-gallon jug contains exactly 2 gallons of water, which satisfies the goal state.

2.Mars Rover – Intelligent Agent

A Mars Rover is an intelligent robotic system that explores the surface of Mars, collects samples, analyzes rocks and soil, and sends scientific data to Earth.

i. What are the percepts of this agent?

Percepts are the information received by the rover through its sensors.

- Camera images of the Martian surface.
- Distance and obstacle information.
- Temperature and environmental conditions.
- Rock and soil composition.
- Rover position, movement, and battery level.

ii. Characterize the operating environment

The Mars Rover operates in an environment with the following characteristics:

- Partially observable: It cannot observe the entire Martian surface.
- Stochastic: Outcomes of movements and experiments may be uncertain.
- Sequential: Each action affects future decisions.
- Dynamic: Environmental conditions may change.
- Continuous: Position, distance, and temperature vary continuously.
- Single-agent: The rover makes its own decisions, although it communicates with Earth.

iii. What actions can the agent take?

- Move forward, backward, and turn.
- Capture images using cameras.

- Collect rock and soil samples.
- Analyze samples using scientific instruments.
- Avoid obstacles and select safe paths.
- Send collected data to Earth.
- Recharge or conserve battery power when possible.

iv. How can we evaluate its performance?

The rover's performance can be evaluated by:

- Number of useful samples collected and analyzed.
- Accuracy of scientific measurements.
- Distance explored successfully.
- Amount of useful data transmitted to Earth.
- Battery efficiency and mission duration.
- Safety and avoidance of damage.

v. Suitable agent architecture and reason

A model-based, goal-based agent with utility-based decision-making is most suitable.It maintains an internal model of its surroundings, selects actions to achieve scientific goals, and compares possible actions based on safety, energy consumption, and scientific value. This helps the rover make decisions even when the environment is uncertain.

3.Eight-Queens Problem Using Search Strategies

Problem: Place eight queens on an \(8 \times 8\) chessboard so that no two queens attack each other.

Problem formulation

- Initial state: An empty chessboard.
- State space: All possible arrangements of queens on the board.
- Actions: Place a queen in a safe square.
- Goal test: Eight queens are placed without sharing the same row, column, or diagonal.
- Path cost: Each placement can have a cost of 1; a solution requires eight placements.

Constraints

1. No two queens can be in the same row.
2. No two queens can be in the same column.
3. No two queens can be on the same diagonal.

Search strategy: Backtracking

Backtracking places one queen in each column. If a queen attacks another queen, the algorithm removes the last queen and tries a different square.

Pseudocode

```
N = 8

Solve(column):
   If column == N:
      Print solution
      Return true

   For each row from 0 to N-1:
      If position is safe:
         Place queen at (row, column)

         If Solve(column + 1):
            Return true  Remove queen


Return false
```

One valid solution

The following row positions represent the queen placed in each column, from column 1 to column 8:

[1, 5, 8, 6, 3, 7, 2, 4]

Column	Queen's row
1	1
2	5
3	8
4	6
5	3
6	7
7	2
8	4

This arrangement satisfies all three constraints, so it is a valid solution.
Conclusion: Backtracking efficiently explores possible placements and rejects invalid configurations before completing the board.

4. OLA Cab Booking – Problem-Solving Agent

A customer uses the OLA application to travel from a source location to a destination and selects a cab type according to comfort, price, and availability.

Type of agent

A goal-based agent with utility-based decision-making is suitable.

- Goal-based: Reaches the customer's destination.
- Utility-based: Compares fare, waiting time, comfort, and cab availability to select the most suitable option.
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Problem formulation

- Initial state: Customer enters the source and destination.
- Goal state: Customer reaches the destination successfully.
- Actions: Enter locations, select cab type, book, pay, and track the ride.
- Constraints: Cab availability, fare, travel time, and customer preference.

Pseudocode

```
START

Input source
Input destination
Input preferred cab type

Find available cabs
Calculate fare and travel time

Display available cab options

If preferred cab is available:
   Select preferred cab
Else:
   Display other available cabs
   Select another cab

Confirm booking
Assign driver
Track cab and ride

If destination is reached:
   Complete ride
   Display final fare
Else:
   Handle ride problem

STOP
```

Performance evaluation

The agent's performance can be measured by successful arrival, minimum reasonable travel time, customer satisfaction, affordable fare, and availability of the selected cab.

Conclusion: A goal-based and utility-based agent helps the customer reach the destination while considering cost, comfort, and convenience.

5. Uniform Cost Search (UCS) Algorithm

A logistics company wants to transport packages from the starting warehouse S to the destination warehouse G at minimum cost. Each route has a cost representing fuel and travel time. The Uniform Cost Search (UCS) algorithm finds the least-cost path by expanding the node with the lowest cumulative path cost first.

Given Weighted Graph

The routes and their costs are:

S → A = 1

- S → G = 12
- A → B = 3
- A → C = 1
- B → D = 3
- C → D = 1
- C → G = 2
- D → G = 3

Steps of UCS

Step	Node expanded	Path considered	Cumulative cost
1	S	S → A	1
2	A	S → A → C	2
3	C	S → A → C → D	3
4	D	S → A → C → G	4
5	G	Goal reached	4

UCS always selects the node with the lowest cumulative cost. The direct path S → G has a cost of 12, which is higher than the indirect routes.

Cost Comparison

Path	Total cost
S → G	12
S → A → B → D → G	10
S → A → C → D → G	6
S → A → C → G	4

Final Result

The Uniform Cost Search algorithm finds the least-cost path S → A → C → G, with a total cost of 4 units. This route is optimal because its cumulative cost is lower than that of all other available paths. UCS guarantees an optimal solution when all route costs are non-negative.